

Nishath Ahamed

Associate Software Engineer | Backend, GenAI and Agentic AI

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PROFESSIONAL SUMMARY

Associate Software Engineer with 2+ years of experience developing backend systems and AI-powered applications for enterprise and fintech environments. Experienced in Java/Spring Boot, Python/FastAPI, SQL, Kafka, RabbitMQ, Docker, Kubernetes, and AWS. Specialized in building GenAI solutions including RAG systems, agent workflows, and LLM-integrated applications using LangChain, LangGraph, and LangSmith. Strong interest in scalable AI infrastructure, backend architecture, and intelligent automation systems.

CORE COMPETENCIES

- **Languages:** Python, Java, TypeScript, JavaScript
- **Frameworks:** Spring Boot, Spring WebFlux, FastAPI, NestJS
- **AI & Agents:** LangChain, LangGraph, LangSmith, OpenAI API, RAG, tool calling, prompt engineering
- **Data & Messaging:** PostgreSQL, MySQL, Chroma, message brokers such as Kafka and RabbitMQ, Prisma ORM
- **Cloud, Containers & Platform:** AWS, Docker, Kubernetes, Jenkins, ArgoCD, Git, Postman
- **Ways of Working:** Agile/Scrum, mentoring, technical writing, cross-functional collaboration

PROFESSIONAL EXPERIENCE

Associate Software Engineer

May 2025 - Present

Axiata Digital Labs, Sri Lanka

Backend engineering for fintech lending and savings platforms, with additional contribution to enterprise GenAI initiatives.

- Build and maintain backend services that support lending and savings workflows across loan creation, retrieval, auto-collection, settlement, and cancellation flows.
- Design and integrate APIs across business-critical loan lifecycle stages while improving maintainability through refactoring and service decomposition.
- Optimize SQL queries and service behavior to improve response times, stability, and operational visibility in production-facing systems.
- Implement event-driven workflows using Kafka and webhook integrations to support automation, asynchronous processing, and external system coordination.
- Contribute to CI/CD delivery and release readiness in collaboration with QA, product, and platform teams.
- Mentor junior engineers during debugging, implementation reviews, and day-to-day delivery work.

GenAI / AI contribution highlights

- Co-designed a Retrieval Augmented Generation subsystem for enterprise core-banking use cases.
- Developed document ingestion and retrieval pipelines covering normalization, chunking, embeddings, and vector search.
- Used LangChain to orchestrate LLM execution and exposed backend functionality through REST APIs aligned with security and compliance expectations.

AI Application Engineer

Sep 2024 - May 2025

Self-employed

Developed RAG systems, chatbots, and multi-agent systems for freelance AI application work.

- Built AI applications using LangChain, LangGraph, LangSmith, vector databases, FastAPI, AWS, AWS Bedrock, and PostgreSQL.

Axiata Digital Labs, Sri Lanka

Supported delivery of backend modules for fintech lending systems and assisted with deployment-oriented engineering work.

- Worked on backend development and integration tasks for loan processing modules, including core-banking-related services and eKYC flows.
- Improved API response times by profiling bottlenecks, refining queries, and assisting with code-level refactoring.
- Helped containerize services and supported deployments using Docker and Kubernetes.

SELECTED PROJECTS

Data Synthesizer | [Axiata Digital Labs](#)

- Designed enterprise-oriented RAG ingestion and retrieval pipelines with semantic-aware chunking, embeddings, vector search, and LangChain orchestration.
- Worked on practical backend patterns for preparing structured and unstructured content for downstream AI retrieval use cases.
- **Tech:** Python, LangChain, Chroma

Decap Hire Purchase | [Axiata Digital Labs](#)

- Implemented APIs and workflow support for automated collections, early settlement, and cancellation processes.
- Integrated event-driven behavior to improve automation across fintech service flows.
- **Tech:** Spring Reactive, Kafka, PostgreSQL, Mambu, Jumio, Splunk, Jenkins, ArgoCD

Boost SME | [Axiata Digital Labs](#)

- Delivered onboarding and revolving-credit-related APIs and supported eKYC integration work.
- Resolved service issues affecting statement-related behavior and backend stability.
- **Tech:** Spring Reactive, Kafka, PostgreSQL, Mambu

AxoCapital SaaS | [Axiata Digital Labs](#)

- Worked on multitenancy and tenant-isolation features for a SaaS platform.
- Contributed to centralized authentication and audit-oriented backend capabilities.
- **Tech:** Spring Boot, Keycloak, PostgreSQL

Multi-agent HR AI Assistant | [Personal Project](#)

- Built a multi-agent HR assistant using OpenAI, LangChain, FastAPI, and PostgreSQL.
- Implemented safe read-only SQL access, supervisor-based routing, clarification handling, and database-backed answer generation through tool calling.
- **Tech:** OpenAI, LangChain, FastAPI, PostgreSQL

Blog Generator ReAct-type Agent | [Personal Project](#)

- Implemented a LangGraph-based ReAct workflow using Groq for blog generation with explicit state transitions and deterministic tool execution.
- Added bounded recursion, intermediate reasoning management, and full execution tracing through LangSmith.
- **Tech:** LangGraph, Groq, LangSmith, Python

AI-Powered Reddit Moderation | [University of Moratuwa](#)

- Created a BERT-based classifier for off-topic and inappropriate content detection.
- Combined language modeling with novelty-oriented features for moderation-related predictions.
- **Tech:** Python, BERT, scikit-learn

CERTIFICATIONS

Quickstart: LangSmith Essentials | [LangChain Academy](#)

Issued Apr 2026 | Credential ID: vd55h4nxdp

- Focused on core LangSmith workflows for tracing, evaluation, and continuous improvement using live production data.
- Strengthened practical understanding of observability and feedback loops for agent engineering.

Quickstart: LangSmith Agent Builder | [LangChain Academy](#)

Issued Mar 2026 | Credential ID: as5soeqwvp

- Covered building, debugging, and evaluating AI agents using LangSmith.
- Included tracing, tool-call inspection, reasoning-path debugging, experiment tracking, and output evaluation.

Project: Ambient Agents with LangGraph | [LangChain Academy](#)

Issued Apr 2026 | Credential ID: lbrbpxlqdp

- Hands-on project on ambient-agent patterns using LangGraph.
- Explored event-driven workflows, human approval flows, state management, tool calling, and long-running agent patterns.

Foundation: Introduction to LangChain - Python | [LangChain Academy](#)

Issued Mar 2026 | Credential ID: jtlromy7bq

- Covered foundational LangChain concepts for building LLM-powered Python applications.
- Included prompt templates, chains, tool integration, memory, and working with multiple LLM providers.

Deep Agents Certification - AI Agent Engineering | [LangChain Academy](#)

Issued Mar 2026 | Credential ID: mao7fnqc8e

- Focused on planning-capable AI systems, long-context reasoning, and multi-agent coordination.

Additional Certificates | [Great Learning Academy](#) / [Simplilearn](#)

Issued 2024

- OOP in Java, Data Structures and Algorithms in Java (Intermediate), Introduction to Database and SQL, and Introduction to Kubernetes.

EDUCATION

- University of Moratuwa
- B.Sc. (Hons) Information Technology
- 2019 - 2025 | Second Class Upper Division

WRITING AND PROFESSIONAL PROFILES

- Regular contributor to Axiata Digital Labs engineering writing and personal Medium posts.
- Writing topics include fintech microservices, API design patterns, RAG integration, Kubernetes deployment, and applied machine learning use cases.
- Notable article: "Building My First Production-Grade RAG System: An Engineer's Entry into Enterprise LLM Integration" (Dec 2025).
- LinkedIn: [linkedin.com/in/ahamed-nishath](https://www.linkedin.com/in/ahamed-nishath)
- GitHub: github.com/nishathmnha
- Medium: medium.com/@nishath.ahamed

LANGUAGES

- Sinhala and Tamil (Native)
- English

REFEREES

- Windana Rahubedde | Principal Architect, Allion Technologies | windanar@alliontechnologies.com | (+94) 11 285 442 / (+94) 76 140 80
- Nipun Chandimali | Lecturer, Department of Information Technology, University of Moratuwa | nipunic@uom.lk | +94 71 079 2617